ACTIVITY PACK

Activity 5: Scenario

Activity 5 - Agent Governance and the Deployment Gate

Course	AI Security for Autonomous AI Agents
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Scenario

Activity 5 - Scenario

Agent Governance and the Deployment Gate

Course: AI Security for Autonomous AI Agents (TGS-2025060473) **Duration:** 25 minutes | **Format:** Small groups of 3 - 4 acting as one governance board | **Type:** Capstone synthetic simulation

Evidence status: SIM. The organisation, people, customer counts, financial values, evaluation results, targets, segment labels, and projected benefits are synthetic classroom data. They are not market evidence or a claim about any real bank or population.

The simulated organisation

Meridian Bank Singapore is a mid-tier retail and SME bank with 1.4 million customers, 26 branches, and a digital-first strategy under pressure from three larger incumbents and two digital banks. It is a Singapore-incorporated entity, regulated by MAS, and is an **organisation** under the PDPA for personal data it collects, uses or discloses. A vendor may separately act as a data intermediary when it processes personal data on the bank's behalf.

Its Collections and Customer Assistance division handles customers who fall behind on unsecured credit - personal loans, credit cards, SME working-capital facilities. The division employs 118 officers and handles roughly **31,000 delinquency cases per year**. Median time from first missed payment to first meaningful customer contact is **nine days**. For this synthetic business case, the bank's commissioned internal analysis assumes that intervention within 72 hours materially improves outcomes for the customer, not only for the bank; it is not presented as external industry evidence.

The proposal

The Chief Digital Officer, **Rachel Tay**, is asking the AI Governance Board to approve production deployment of **"ARIA"** - Assisted Recovery and Intervention Agent.

ARIA is not a chatbot. It is an **autonomous agent**: a planning loop over a commercial frontier model, with tool access and persistent per-customer memory. Its mandate is to detect early delinquency, contact the customer, understand the situation, and **agree and enact a restructuring** without an officer in the loop for standard cases.

What ARIA is permitted to do in the proposed design

#	Tool / action	Reversible?	Visible to customer?
1	Read full customer profile: account history, transactions, employment field, CPF contribution flag, demographic data	n/a	No
2	Read the bank's collections policy and hardship guidelines (RAG over an internal document store)	n/a	No
3	Initiate outbound contact by SMS, email and in-app message	No	Yes
4	Conduct a multi-turn conversation about the customer's financial circumstances	n/a	Yes
5	Offer and **execute** a payment-plan restructuring up to S$8,000 exposure	Partially	Yes
6	Apply a **hardship flag** to the account	Yes, but recorded	Not directly
7	Waive late fees up to S$300 per case	Yes	Yes
8	**Escalate the account to the external debt recovery agency**	**No**	Yes, severely
9	Write a case summary into the CRM, used by human officers and by future ARIA sessions	Persistent	No
10	Submit a data point to the credit bureau reflecting restructuring status	**No**	Consequential

Actions 1 - 9 are proposed as **fully autonomous**. Action 10 is proposed as autonomous "because it is a factual reporting obligation, not a decision."

The business case

Metric	Today	Projected with ARIA	Basis
Median days to first contact	9.0	0.4	Agent runs continuously
Cases handled per year	31,000	31,000	Same volume
Officer FTE required	118	34	Vendor estimate
Annual operating saving	-	S$6.1m	Finance projection
Cure rate (customer returns to good standing)	41%	55% (projected)	Vendor pilot data, different market

Rachel Tay's paper to the board states: *"Every week we delay costs us S$117,000 and leaves customers in distress nine days longer than necessary. The ethical case for ARIA is as strong as the commercial one."*

She is not wrong about the harm of the status quo. That is what makes this decision difficult.

The evaluation results

The bank's model risk team ran a 6,000-case retrospective evaluation, replaying real historical cases through ARIA and comparing its proposed action against the outcome a senior officer panel judged correct. Results were tabled 48 hours before the board meeting.

Overall performance

Metric	Result	Internal target
Recommended-action accuracy (all segments)	91.2%	≥ 90%
Hallucination rate - ARIA asserted a policy provision, fee, or account fact that does not exist	**4.7%**	Not specified
Of those hallucinations, proportion that resulted in a **wrong tool call being executed**	**38%**	Not specified
Policy-adherence violations (offer outside mandated bands)	2.1%	≤ 1%
Median conversation turns to resolution	6	-
Cost per case	S$0.42	-

Accuracy by customer segment

Customer segment	Cases in eval	Accuracy	False "escalate to recovery agency" rate	Median offer generosity index
Singapore citizen, salaried, English primary	2,410	**94.6%**	1.2%	1.00 (baseline)
Singapore citizen, salaried, non-English primary	890	89.1%	3.4%	0.91
Singapore PR, salaried	640	90.8%	2.1%	0.96
Work Permit / S Pass holder	510	**82.3%**	**7.8%**	**0.74**
Self-employed / gig economy income	780	84.9%	6.1%	0.79
SME sole proprietor	470	88.4%	4.0%	0.88
Age 60+	300	86.7%	5.2%	0.93

The model risk lead's note reads, in full:

> "Aggregate accuracy of 91.2% clears our threshold. Segment variance is expected given training data distribution and case-mix differences. Recommend proceed with monitoring."

One further finding, filed as low severity

During the evaluation ARIA generated internal automation scripts to reconcile payment plans. In 11 instances it imported Python packages that **do not exist** - plausible-sounding names assembled from its training distribution. A developer had pip-installed two of them before noticing. Neither resolved. The finding was logged as *"minor code quality issue"*.

What the board already knows

- ARIA's supplier is a well-regarded international vendor. Meridian operates the deployment and, using the PDPC's 2 June 2026 proposed consultation vocabulary, is the **system deployer**.

- The customer notification currently drafted for the terms and conditions reads:

"We may use automated systems and new technologies to improve our products and services."

- No customer-facing disclosure currently states that the counterparty in a collections conversation is an AI agent.
- ARIA's per-customer memory is retained indefinitely, "for continuity of service."
- Conversation transcripts, model prompts and tool-call logs are stored in the vendor's regional cloud tenancy. Nobody has asked which region.
- The bank's audit trail cannot presently distinguish an action taken by ARIA from one taken by a human officer. Both appear in the CRM as "Collections - system."
- A director has asked whether ARIA can be extended next year to **SME credit decisioning**.

Your role

You are the **AI Governance Board of Meridian Bank Singapore**. Your members are the Chief Risk Officer (chair), the Data Protection Officer, the Head of Collections, an independent non-executive director, and the Head of Model Risk. Rachel Tay is presenting; she is not a member and does not vote.

You must return a **written decision** to the Board of Directors. It must contain:

1. **Go / no-go / go-with-conditions** - with the conditions specified precisely enough to be tested.
2. **An autonomy matrix** - for each of ARIA's ten capabilities: *alone*, *with human approval*, or *never*.
3. **A human accountability model** - who is answerable for an ARIA decision, at what checkpoints, and what the override and appeal path is for a customer.
4. **Monitoring and audit requirements** - what is logged, what is measured, what triggers suspension.

You have 25 minutes. Your decision will be read by people who will lose S$117,000 a week if you are too cautious, and who will front a MAS and PDPC enquiry if you are not cautious enough.